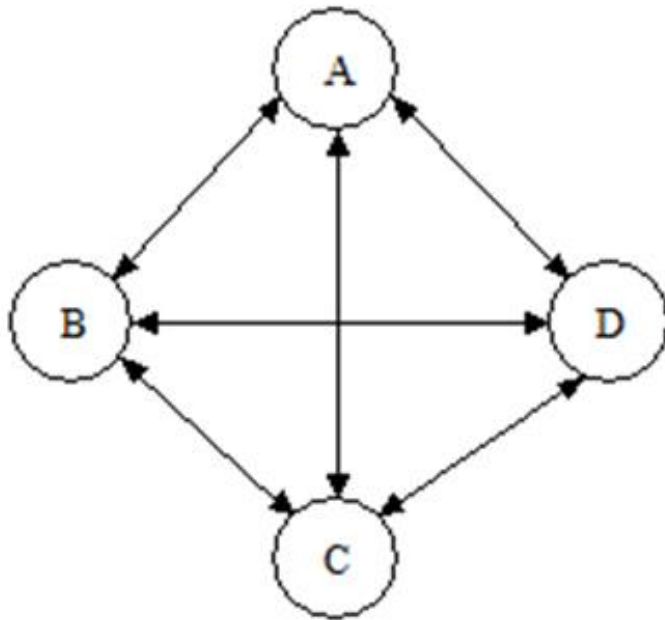


Communication Networks and Protocols

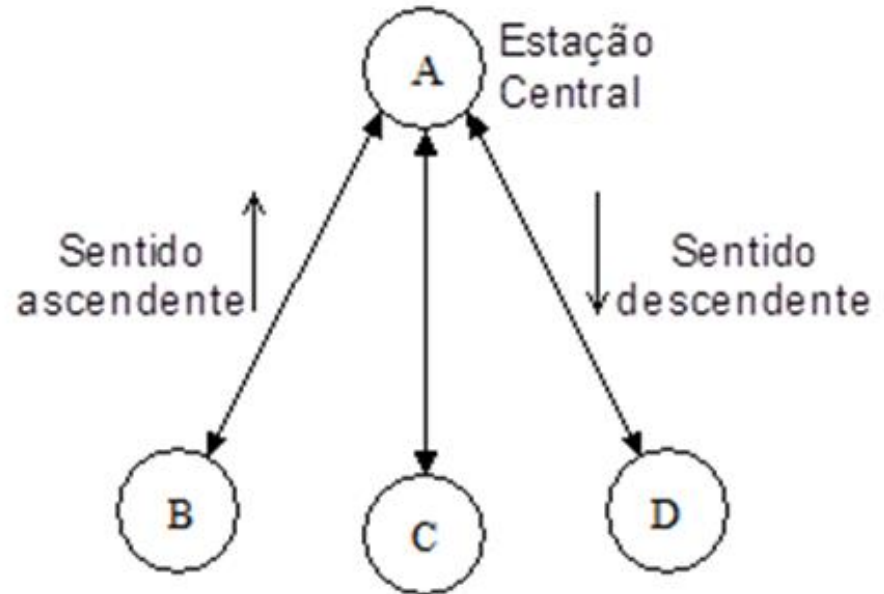
José Augusto Afonso

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Direct vs. Centralized Networks



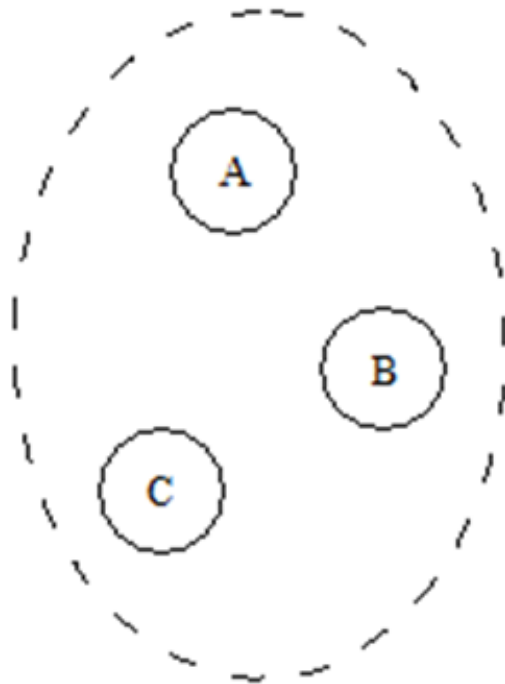
Comunicação directa



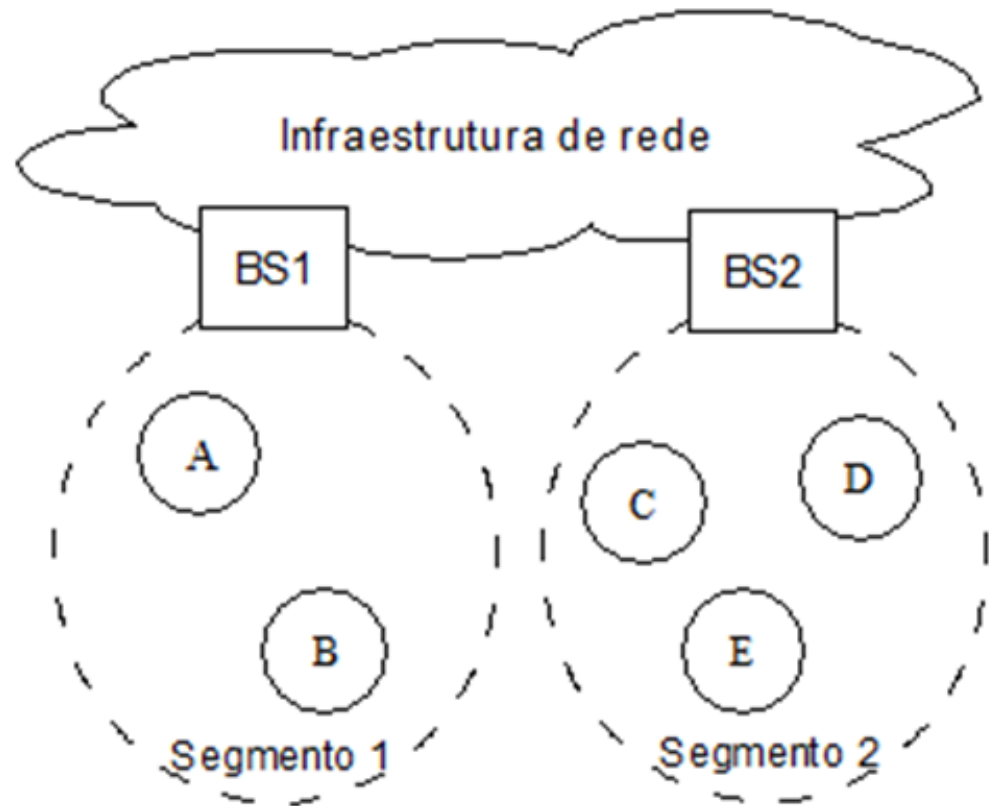
Comunicação centralizada

- Central station: Master, Base station, Access point
- Topologies
 - Single Hop (Peer-to-Peer, Star) vs. Multihop (Mesh, Tree)
 - Physical vs. Logical, Wireless vs Cabled (Bus, Ring, etc.)

Ad hoc vs. Infrastructure-based Networks

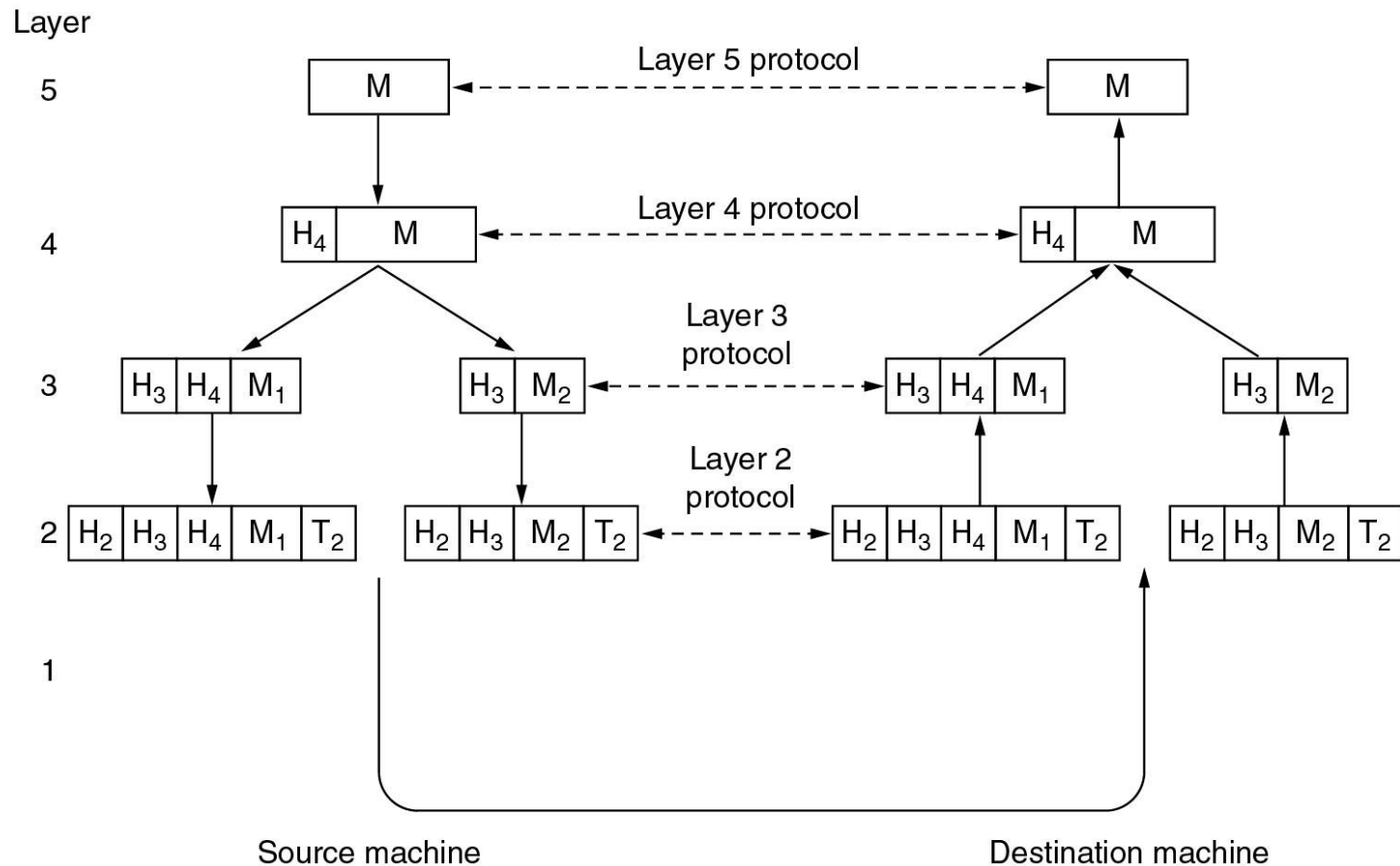


Rede ad hoc



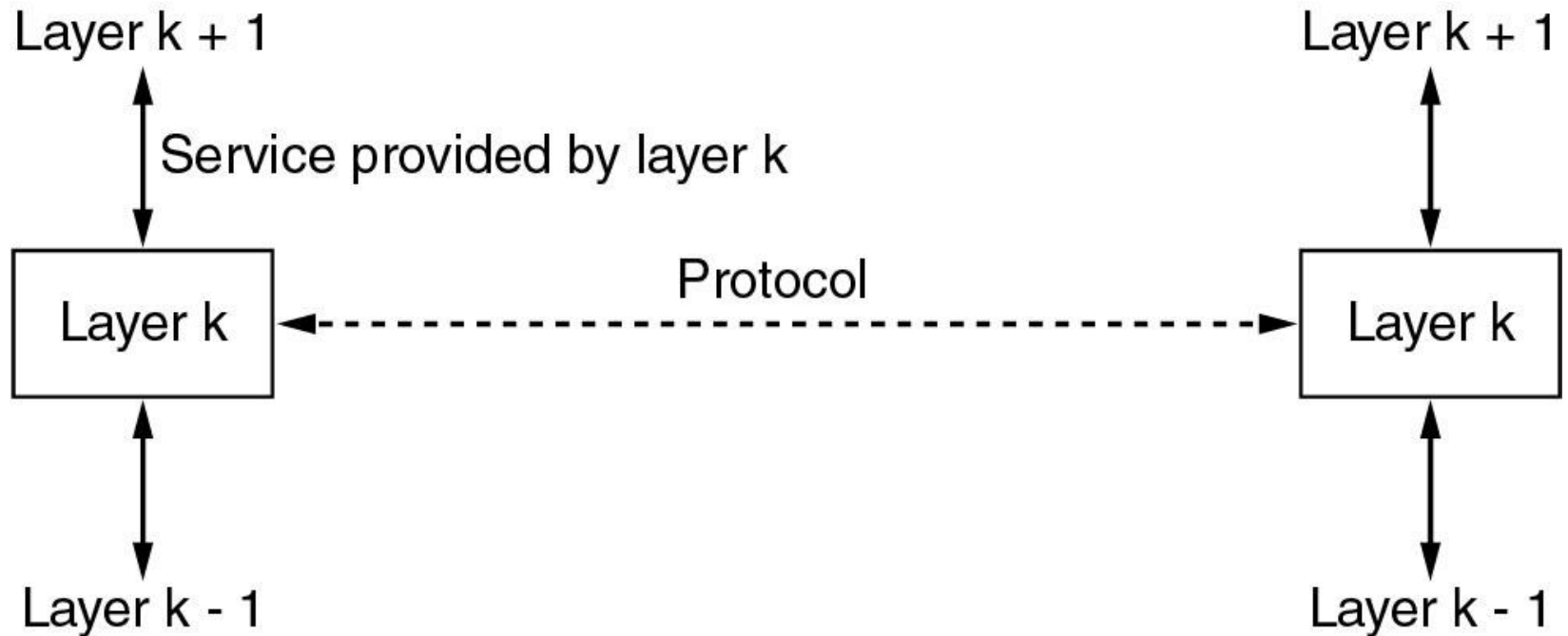
Rede baseada em infraestrutura

Protocol Hierarchies

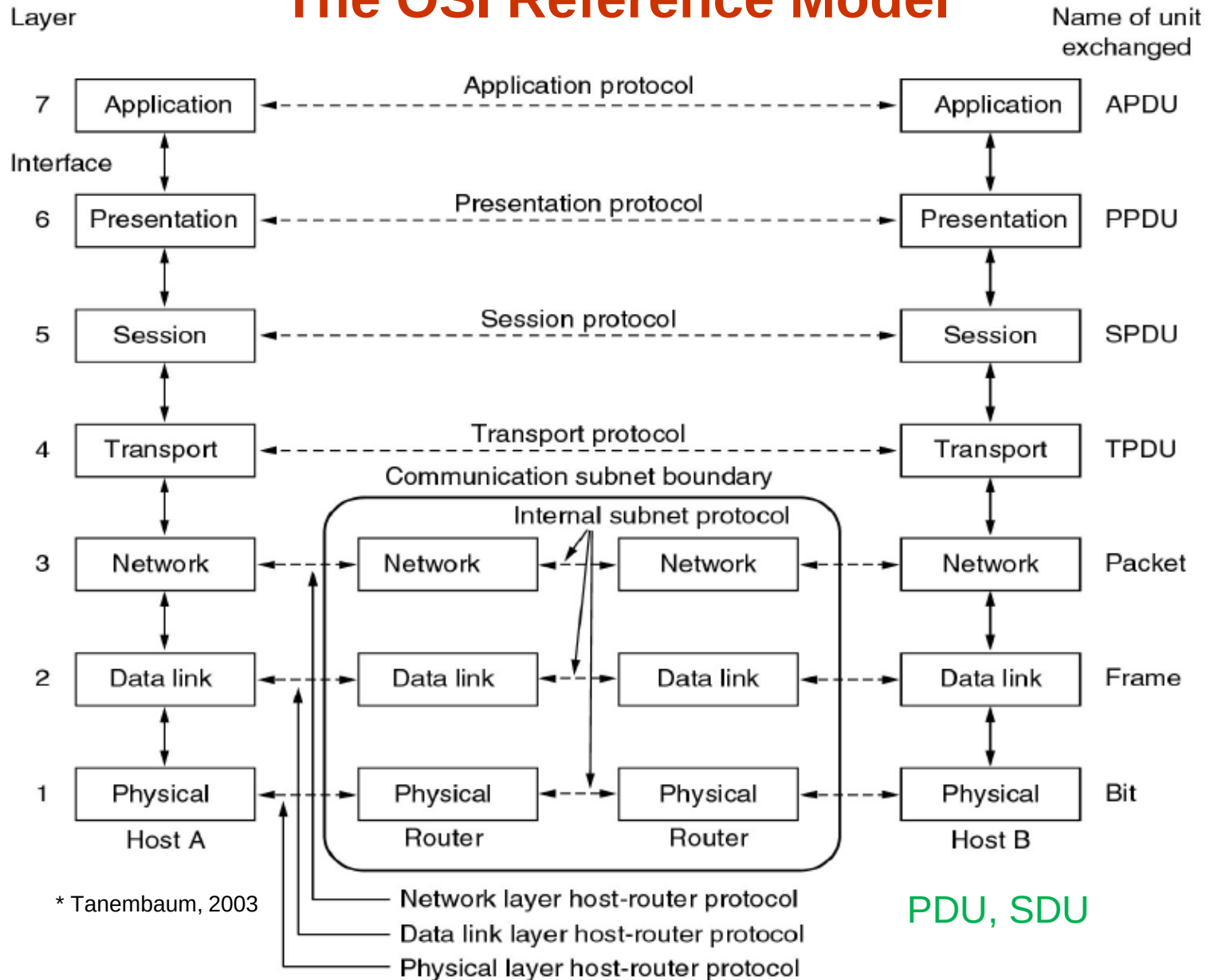


Protocols, Modularity (Layers), Encapsulation, Layer Independence

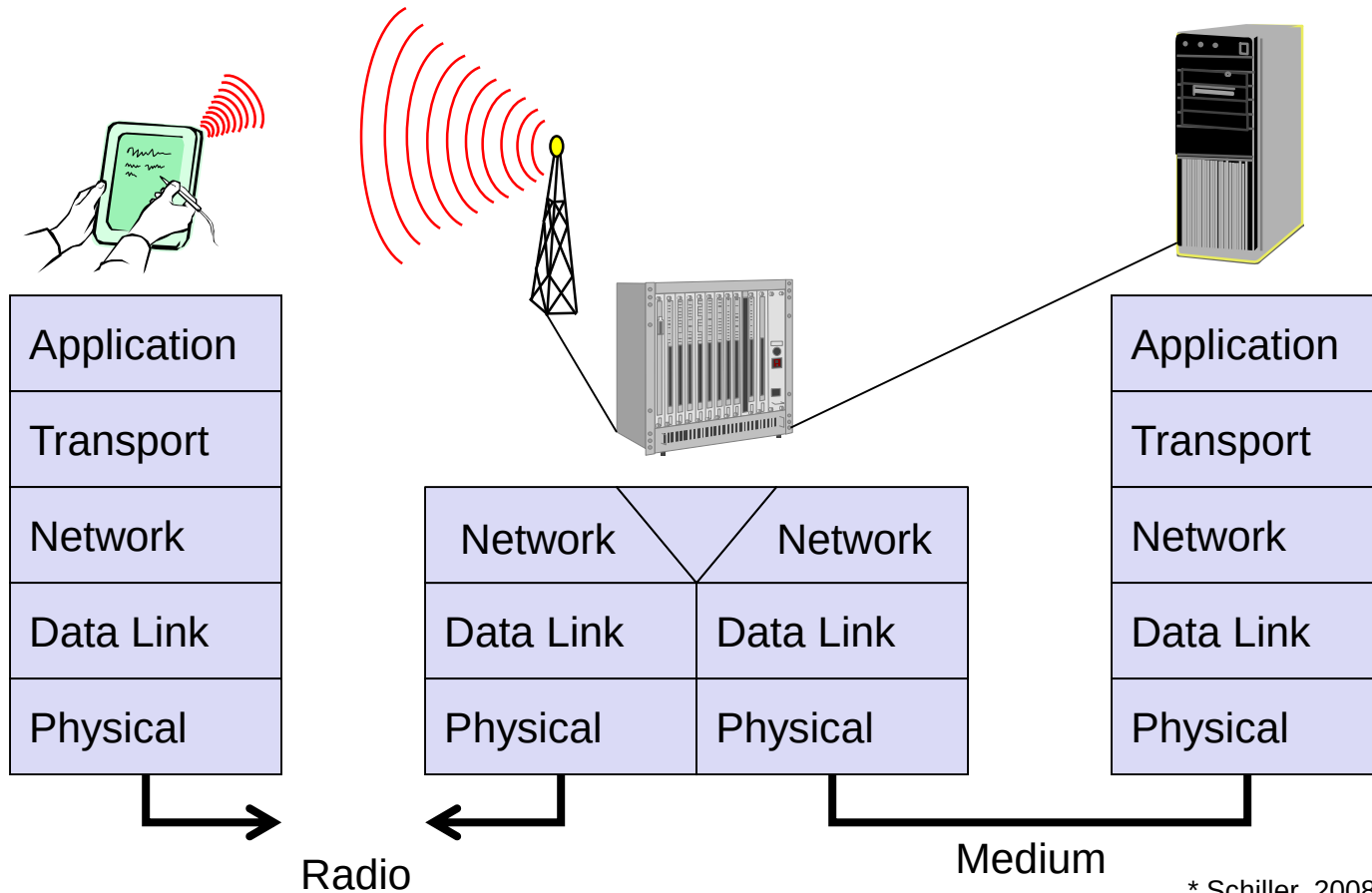
Services to Protocols Relationship



The OSI Reference Model



Reference Model Used in Practice

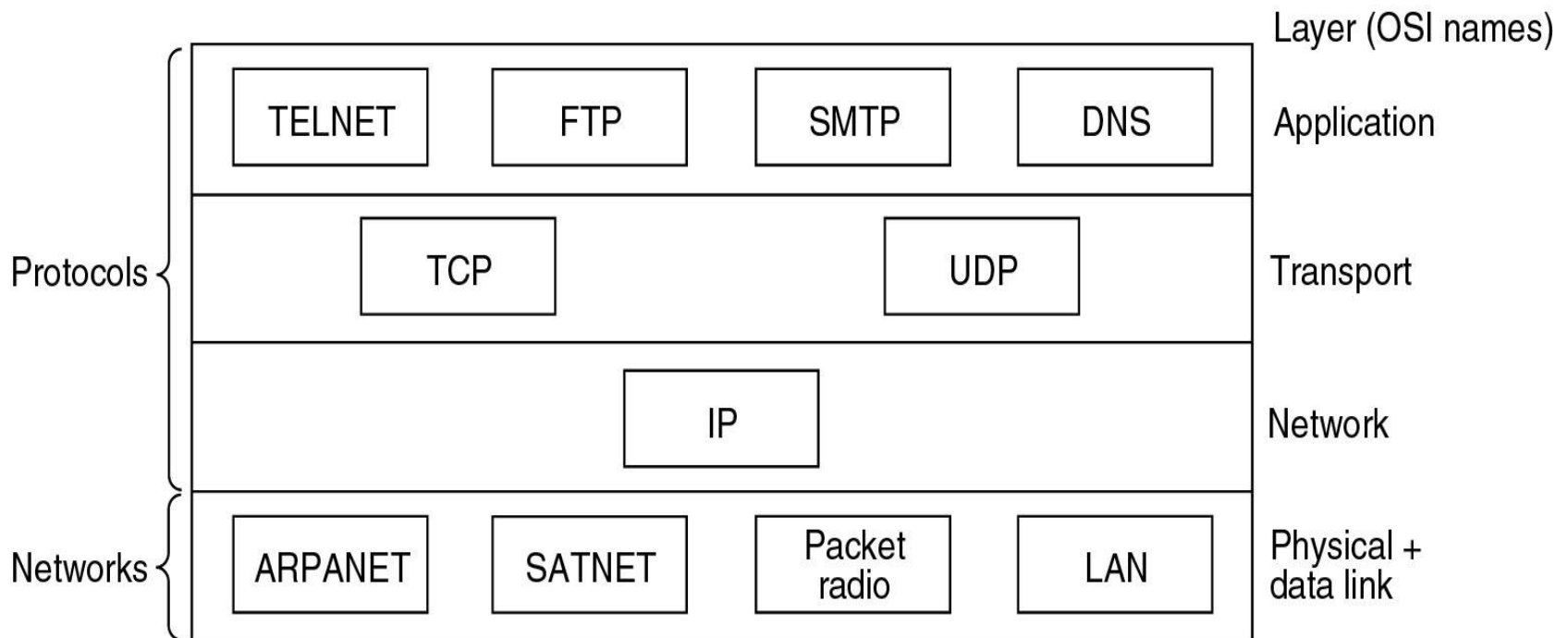


* Schiller, 2008

Internetworking, Gateways

Stack Example: TCP/IP

- Protocols and networks in the TCP/IP model initially.



Functions of the Layers

- **Physical Layer**

- Concerns transmission of the bit stream at the electrical and mechanical level
- Modulation
- Synchronization
- Error Correction

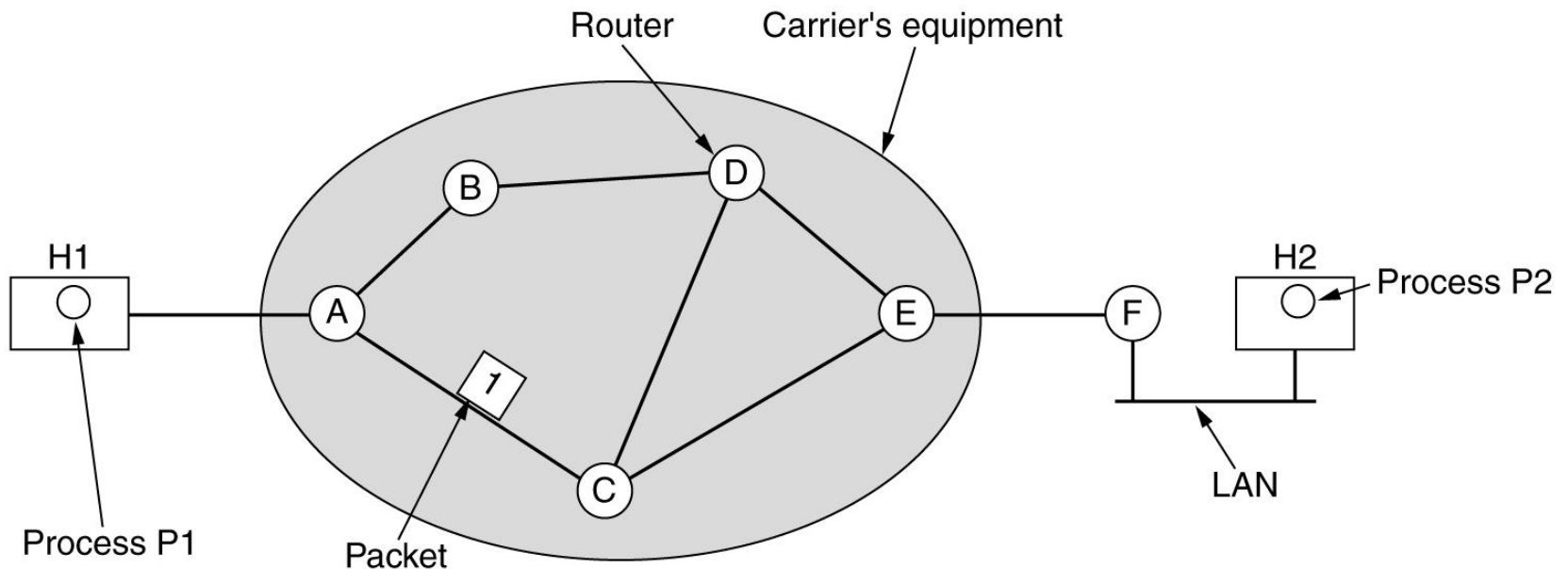
- **Data Link Layer**

- Framing – Identification of start and end of frames
- Error Control
- Flow Control
- Medium Access Control

Functions of the Layers (cont.)

- **Network Layer**

- Routing
- Addressing
- Internetworking
- Congestion control



Functions of the Layers (cont.)

- **Transport Layer**
 - End-to-end error control and flow control
 - Multiplexing
- **Application Layer**
 - Supports application and end-user processes